MIL-T-43578B
21 November 1978
SUPERSEDING
MIL-T-43578A
10 February 1972

MILITARY SPECIFICATION

TROUSER TOP FORMER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a trouser top former for finishing the top of trousers (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CC-M-1807 - Motors, Alternating-Current, Fractional Horsepower and Integral Horsepower (500 Horsepower and Smaller)

MILITARY

MIL-T-152 - Treatment, Moisture-and Fungus-Resistant, of Communications, Electronics, and Associated Electrical Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3510

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MIL-L-3153 - Laundry and Dry Cleaning Machinery and Equipment (For Fixed Installations), Preparation for Delivery Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-1474 - Noise Limits for Army Material

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - Industrial Controls and Systems

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017 l)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

UNDERWRITERS LABORATORIES INC. (UL) STANDARDS

(Application for copies of applicable UL Standards should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, Long Island, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050; or 2602 Tampa East Blvd., Tampa, FL 33619.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. Each trouser top former furnished under this specification shall be the manufacturers current standard product, except for those changes required to comply with the requirements of this specification.

3.2 Codes and standards.

3.2.1 NFPA. The top formers shall be wired to conform to NFPA Standard No. 70.

3.2.2 UL. The wiring, fittings, and conduit shall conform to applicable UL standards.

3.2.3 NEMA. Motor controllers, switches, solenoids and time delays shall conform to performance requirements of NEMA Standards Publication No. ICS.

3.2.4 Certification. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative, that the top formers meet the requirements specified herein.

3.2.4.1 NFPA. Acceptable evidence of meeting requirements of 3.2.1 shall be the manufacturer's certified statement that the top former is wired in accordance with NFPA Standard No. 70.

3.2.4.2 UL. Acceptable evidence of meeting the requirements of 3.2.2 shall be the UL label or a certification from an independent testing laboratory acceptable to the contracting officer or his authorized representative that the wiring, fittings and conduit conform to applicable UL standards.

3.2.4.3 NEMA. Acceptable evidence of meeting the requirements of 3.2.3 shall be the manufacturer's certified statement that the motor controllers, switches, and time delays conform to NEMA Publication ICS. A tag or label attached to these items stating the components conform to this standard is acceptable evidence.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for the specified equipment provided the completed item complies with all provisions of this specification (see 6.3).

3.3.1 Motors. Fractional and integral horsepower motors shall conform to CC-M-1807.

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3.3.2 Conduit, fittings and wire. Electrical conduit shall be rigid electrical, metallic tubing or flexible conduit. All electrical wire shall be stranded copper heat-resistant grade, thermoplastic insulated. Solid (nonstranded) wire shall not be used. All conduit fittings and electric wire shall be UL listed or labeled.

3.4 Design and construction. The machine shall be designed for fully automatic operation. The machine shall have a welded steel frame to support controls, air expanding form, steamer and blower. The frame shall be provided with anchor bolts and removable panels for inspection and maintenance of enclosed mechanisms. The machine shall be provided with a hand operated, moisture spray gun for touching wrinkles. The electrical components shall be designed for operation on the electrical characteristics as specified (see 6.2) . Assemblies of pipes, tubing, fittings, and coils using steam shall have no visible evidence of leakage or failure when tested as specified in 4.3.1. Air lines and cylinders shall have no evidence of leakage or failure when tested as specified in 4.3.2.

3.4.1 Form air expanding. The form shall be a nylon bag that is capable of expanding and conforming to the size and shape of the trouser top and seat. The bag shall contain a perforated or steam-heated padded buck for transfer of steam and air (see 4.3.3).

3.4.2 Plates pleat pressing. The plates shall be double acting and self-aligned to fit the tailored cut of trousers and an air pressure regulator and gauge shall be furnished for the plates (see 4.3.3).

3.4.3 Expander waist band. The waist band expander shall be designed in accordance to the manufacturer’s commercial practice to exert pressure on the entire inside of the waist band of the trousers being finished (see 4.3.3).

3.4.4 Clamp waist band. The waist band clamp shall be an air or spring actuated clamp lock with pushbutton or pedal controlled release (see 4.3.3).

3.4.5 Blower. The blower shall be equipped with a fan to drive air across the steam coil and into the air expanding bag (see 4.3.3).

3.4.6 Controls. The machine shall be equipped with adjustable timers for automatic control of the flow of steam and hot air into the bag form. The controls shall also automatically release the waist band expander (see 4.3.3).

3.5 Performance. The unit shall conform to the following requirements when tested as specified in 4.3.4.

(a) Pleat plates, and waist band expanders are in proper adjustment.
(b) No leaks.
(c) Compatability of the waist band expander for 28 inch (711 mm) and 48 inch (1220 mm) trouser measurements.

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3.6 Marking for identification. Marking for identification shall be in accordance with the manufacturer's commercial practice.

3.7 Finish and color. The machine shall be finished in accordance with the manufacturer's commercial practice. Stop pushbuttons shall be red.

3.8 Noise limits. The machine shall meet the noise limits specified in MIL-STD-1474 and where practicable, shall not generate 85 decibels A-weighted or more at the normal operating position (see 4.3.5).

3.9 Electromagnetic compatibility. When specified (see 6.2), equipment shall be designed and equipped to conform to class IIB of MIL-STD-461 (see 4.3.6).

3.10 Moisture and fungus resistant treatment. When specified (see 6.2), electrical components shall be moisture and fungus resistant treated in accordance with MIL-T-152.

3.11 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.4).

3.12 Government-furnished property. Trousers will be furnished by the Government (see 6.2) for test as specified in 4.3.4.

3.13 Workmanship. The machine shall be free from defects such as fractures, splits, punctures, tears, dents, creases, deteriorations, or malformations. There shall be no sharp edges, burrs, or slivers. The finish applied to the end item shall be continuous, smooth, adherent, have no imbedded foreign matter and contain no sags, runs, drips, creeps, laps, bubbles or streaks. No rust, or tool marks shall show through the coating. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits shall be removed from the welds. Threaded fasteners shall not be broken, stripped, fractured, or loose. Rivets shall be driven to completely fill the holes, with full size heads neatly finished and in full contact with the surface of the riveted members. Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall have adequate slack to provide strain relief.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. A lot shall be all trouser top formers offered for inspection at one time. The sample unit shall be one former.

4.2.2.1 Visual examination. The completely assembled end item shall be examined for defects listed in table 1. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 for major and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Finish not in accordance with the manufacturer's commercial practice	X	
	Not continuous, smooth, adherent, foreign matter imbedded, and sags, runs, drips, creeps, laps, bubbles, or streaks		
	Rust, or tool marks		

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Design and construction and workmanship, general (applicable to all components and assemblies)	Any part missing or not specified type	X	
	Sharp edge, burr, or sliver	X	
	Split, punctured, torn, dented, deteriorated or malformed	X	
Marking	Not manufacturer's commercial practice		X
Welding and brazing	Missing, incomplete, burn holes, fractured, or not fused	X	
	Slag inclusion, undercut, not smooth and uniform, or deposit of scale and flux not removed	X	
Bolts, nuts, screws, studs, and other threaded fasteners	Missing, broken, stripped, or fractured	X	
	Threaded fastener loose		X
Rivets	Missing, broken, not peened or not peened sufficiently	X	
	Rivets loose		X
Electrical apparatus wiring	Wiring cut, abraded, insulation stripped, not properly joined or loose at terminals	X	
	Not inclosed in specified conduit where required	X	
	Adequate slack not provided for wiring to relieve strain		X
Marking for identification	Missing, incomplete, not legible		X

4.2.2.2 Testing of the end item. The first produced end item shall be tested as specified in 4.3.1 through 4.3.5 and, when applicable 4.3.6. Remaining end items of the same lot shall be tested as specified in 4.3.1 and 4.3.2. Failure to pass tests shall be cause to reject the effected end item.

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4.2.3 Code and standards compliance. Proof of compliance with the requirements of 3.2 shall be made available to the Government representative. When moisture and fungus resistant treatment is specified, the contractor shall furnis a certificate of compliance that electrical components comply with 3.9.

4.2.4 Inspection of preparation for delivery. Preservation-packaging and marking shall be inspected with the quality assurance provisions of MIL-L-3153.

4.3 Tests.

4.3.1 Hydrostatic pressure tests. Assemblies of pipes, tubing, fittings and coils using steam shall be tested for 10 minutes with a 200 pound per square inch gage (1380 kPa) hydrostatic pressure. Any evidence of leakage or failure (see 3.4) shall constitute failure of this test.

4.3.2 Air pressure tests. All air lines and cylinders shall be tested for 10 minutes with 80 psig (552 kPa) air pressure. Any evidence of leakage or failure (see 3.4) shall constitute failure of this test.

4.3.3 Operational test. Each unit shall be connected to electric power, air and steam and operational tested without trousers. Nonconformance with the requirements of 3.4.1 to 3.4.6 shall constitute failure of this test.

4.3.4 Functional test. The unit shall be connected to electric power, and steam, and functionally tested under full load requirements with GFP trousers for conformance with 3.5. During this test the unit shall be checked for 4.3.3. Nonconformance shall constitute failure of this test.

4.3.5 Noise test. The unit shall be tested for conformance with the noise limits of paragraph 3.8 in accordance with the applicable procedures of MIL-STD-1474.

4.3.6 Electromagnetic compatibility. When electromagnetic compatibility is required, the initial unit of production shall be tested by the contractor in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting office written certification that the Interference Control Plan, the ENI/FMC Test Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461 (see 3.9).

5. PACKAGING

5.1 Preservation-packaging, packing and marking. Preservation-packaging, packing and marking for level A, B, or Commercial shall be in accordance with MIL-L-3153 (see 6.2).

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6. NOTES

6.1 Intended use. The trouser top former covered by this specification is intended to be used in fixed laundry and dry cleaning plants for finishing the tops of men's laundered polyester or dry cleaned trousers (particularly of the durable press type) in one operation.

6.2 Ordering data. Procurement documents should specify the following, as applicable:

(a) Title, number, and date of this specification.
(b) Required electrical characteristics (see 3.4).
(c) When electromagnetic compatibility is required (see 3.9).
(d) When moisture and fungus resistant treatment is required (see 3.10).
(e) Contract data requirements (see 3.11).
(f) Source of Government-furnished property (see 3.12 and 6.5).
(g) Selection of applicable levels of preservation-packaging and packing (see 5.1).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.4 Contract data requirements. Any data items to be delivered under any contract for items covered by the specification shall be specifically called for in the contract in accordance with the applicable regulations of the procuring agency (see 3.11 and 6.2).

6.5 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.12.

6.6 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-43578B Trouser Top Former

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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